ThermoX ONE BOARD: COPPER (82 mm round, ESP32 rows 22.86 mm)

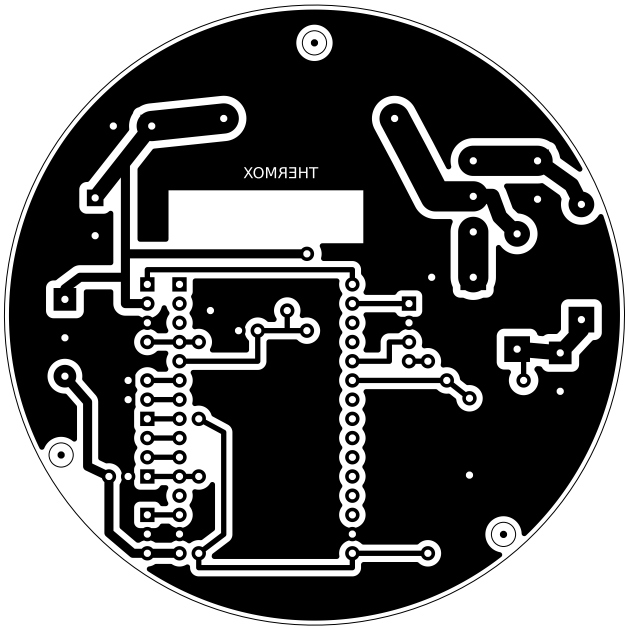
Print at 100 % / Actual size (no 'fit to page'). Check the 50 mm bar with a ruler.

Use THIS page only if your ESP32 pin rows are 22.86 mm apart (centre to centre). Otherwise use the other PDF.

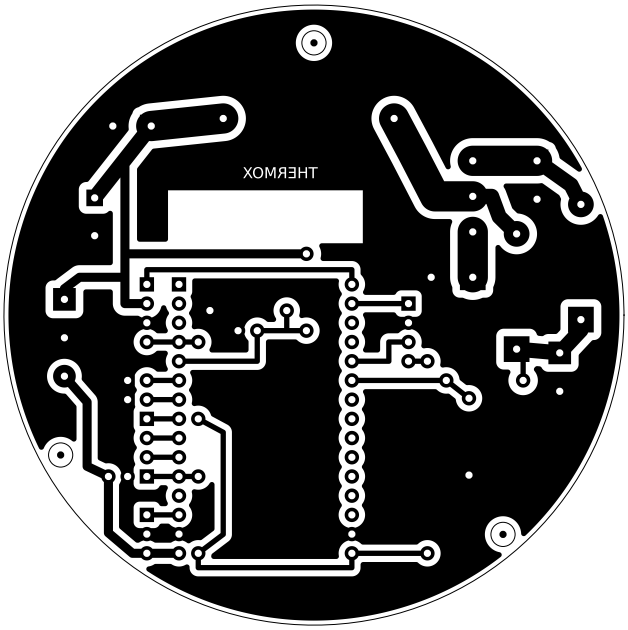
Toner transfer or UV film: use this page as it is (do not mirror). Copper is on the bottom of the board.

Check: the small text 'THERMOX' looks mirrored on the paper and reads normally on the etched copper side.

Drill: 2.2 mm SW1, 1.5 mm F1 and M1 wire pads, 1.3 mm screw terminals and U2, 1.0 mm C1, ESP32 socket and headers, 0.8 mm resistors and C4, 3.2 mm mounting holes. Cover the wide tracks with solder.



copy 1

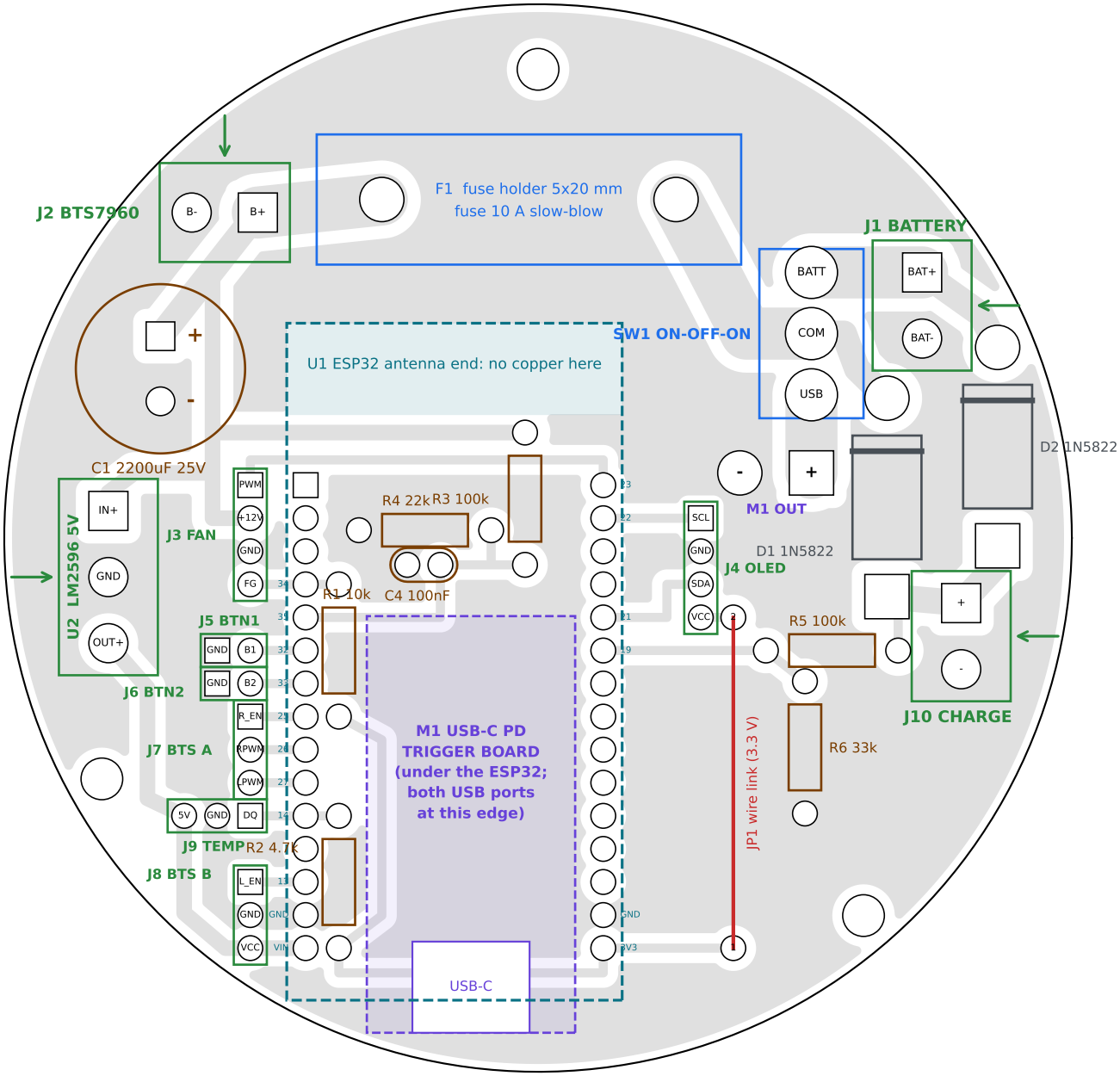


copy 2 (spare)

50 mm: measure this after printing. It must be exactly 50 mm.

ThermoX ONE BOARD: PARTS SIDE (top view, enlarged 2 x)

Parts go on this side, copper is underneath (grey). Square pad = pin 1 (+ on C1, IN+ on U2).
Order: R1-R4 and C4 (they lie flat under the ESP32), M1 (dashed purple, under the ESP32), then the rest,
the two 1x15 female headers for the ESP32 last. Plug the ESP32 in with its USB end at the board edge.
M1: set it to 12 V and measure first. Stick it down with thin foam tape, USB-C at the edge, and run its 2 output
wires (1 mm2) out under the ESP32 antenna end to the M1 + and - pads. M1 must be 7 mm tall or less.
Arrows show where the wires go into the screw terminals. D1, D2: the stripe (cathode) as drawn. Step by step: page 4.



ThermoX ONE BOARD: WIRING, PARTS, NOTES

CONNECTIONS

J1 BATTERY	BAT+: battery + after the BMS (P+). BAT-: battery - (BMS P-).
J2 BTS7960	B+ and B- to the BTS7960 power terminals (1.5 mm2 wire). M+/M- of the BTS7960 go to the Peltier.
J3 FAN	SUNON fan: PWM blue, +12V red, GND black, FG yellow (tach).
J4 OLED	1.5" SH1107 OLED: SCL, GND, SDA, VCC (3.3 V). Match the names on the OLED.
J5 / J6	Button 1 (DOWN) / Button 2 (UP): one leg to B1/B2, the other to GND.
J7 BTS A	BTS7960 control: R_EN, RPWM, LPWM.
J8 BTS B	BTS7960 control: L_EN, GND, VCC (5 V logic). R_IS and L_IS stay open.
J9 TEMP	DS18B20: DQ yellow, GND black, 5V red (R2 4.7k pulls DQ up to 3.3 V).
U2 LM2596 5V	LM2596S buck module (24V/12V to 5V): IN+ to module IN+, GND to IN- (and OUT-), OUT+ to module OUT+. The module goes beside the board (foam tape); use 0.5 mm2 wires, 10-15 cm max.
M1 (on board)	USB-C PD trigger board under the ESP32, USB-C at the board edge. Output + / - to the M1 pads.
SW1 (on board)	ON-OFF-ON: BATT = battery, middle = off, USB = trigger board. Holes also take 3 wires to a case switch.
J10 CHARGE	DC jack 5.5 x 2.1 mm on the case: centre pin (+) to J10 +, sleeve (-) to J10 -. Check with a meter.

IMPORTANT

Only ONE source at a time: SW1 picks battery or USB-C. Never connect the trigger board straight to the battery.
Set the trigger board to 12 V and measure it BEFORE fitting it. Never 15 V or 20 V. On USB-C keep the Peltier at 60 % (12V 5A charger) or 40 % (12V 3A) with the phone page slider.
SW1 (MTS-103) is rated 6 A: on battery keep the Peltier at 80 % or less, or use a 10 A toggle on wires.
Battery gauge: R3 100k / R4 22k on GPIO35. ThermoX.ino already has BATT_CELLS = 3 and BATT_DIVIDER = 5.545; to show battery % set ENABLE_BATTERY_SENSE = true. On USB-C power it shows the USB voltage, not the battery.
Fan tach (FG) is on GPIO34 as before. All other pins are the same as the firmware.
LM2596S module: if it has a trimmer, set it to 5.0 V with a meter BEFORE wiring it to U2. Never more than 5.2 V.
Before the ESP32 goes in: check that BAT+, PD+, COM, BUS, 5V and 3V3 are not shorted to GND. Switch on BATT and measure 5.0 V between the ESP32 socket VIN and GND pins. Only then plug in the ESP32.
Charging: 12.6 V Li-ion charger (3S, CC/CV) into J10 only. D2 charges the battery (to about 90 %), D1 powers the board, R5/R6 tell GPIO19. The OLED shows CHARGING and ThermoX will not run until the charger is unplugged.
Never plug the USB-C charger (trigger board) and the DC charger in at the same time.

PARTS

U1: ESP32 DevKit 30-pin + two 1x15 female headers (8.5 mm tall) M1: USB-C PD/QC trigger board, max 32 x 16 x 7 mm
U2: LM2596S 5 V buck module (you have it) + 3 wires 0.5 mm2, or a 3-pin screw terminal 5.08 mm (KF301-3P)
SW1: MTS-103 ON-OFF-ON F1: BLX-A 5x20 holder + 10 A slow-blow fuse C1: 2200 uF 25 V low ESR, max 13 mm
R1 10k, R2 4.7k, R3 100k, R4 22k (1/4 W) C4: 100 nF ceramic J1, J2: 2-pin screw terminal 5.08 mm (KF301)
J3 4-pin, J4 4-pin, J5/J6 2-pin, J7 3-pin, J8 3-pin, J9 3-pin male headers 2.54 mm M3 screws: 3
Charging: J10 2-pin screw terminal (KF301) D1, D2: 1N5822 (3 A Schottky) R5 100k, R6 33k DC jack 5.5x2.1 (female)
+ 12.6 V 2 A Li-ion charger (3S). NOT a 21 V (5S) charger.

DESIGN CHECK (run by make_thermox_pcb.py)

clearance >= 0.6 mm between nets, all nets connected, no copper under the ESP32 antenna, ground path for the Peltier current >= 3.2 mm wide, no part bodies overlapping: OK

ThermoX ONE BOARD: TUTORIAL (build, test, charge)

A. MAKE THE BOARD

1. Measure your ESP32: centre of a left pin to centre of the right pin, same row. 25.4 mm or 22.86 mm: use that PDF.
2. Print page 1 at 100 % (Actual size) on glossy paper with a laser printer. The 50 mm bar must be 50 mm.
3. Lay the parts on the paper: ESP32, fuse holder, switch, D1, D2, screw terminals. Check the holes line up.
4. Toner transfer, etch, remove the toner, drill (sizes on page 1). The text THERMOX reads normally on the copper.
5. With a meter (beep), check that no two neighbouring tracks touch.

B. SOLDER (low parts first)

6. Under the ESP32, flat on the board: R1 10k, R2 4.7k, R3 100k, R4 22k, C4 100 nF.
7. JP1 wire link, R5 100k, R6 33k, D1 and D2 1N5822: the stripe on the diode goes to the stripe on page 2.
8. Headers J3-J9, screw terminals J1, J2, J10, U2 wires (or KF301-3P), F1 holder, C1 (stripe = -), SW1.
9. M1 trigger board: set it to 12 V and measure. Foam tape under it, USB-C at the edge, 2 wires to M1 + and -.
10. The two female headers for the ESP32 last. Do NOT plug in the ESP32 yet.

C. FIRST POWER-UP (no ESP32, no BTS7960)

11. Meter: BAT+, PD+, COM, BUS, CHG, 5V and 3V3 must NOT beep to GND.
12. LM2596S module: set it to 5.0 V with a meter, then wire it to U2 (IN+, GND, OUT+).
13. Battery on J1, SW1 on BATT: ESP32 socket VIN = 5.0 V, J2 B+ = battery voltage. SW1 middle: everything 0 V.
14. SW1 middle, charger on J10: VIN = 5.0 V (the charger powers the board) and the battery voltage slowly rises.

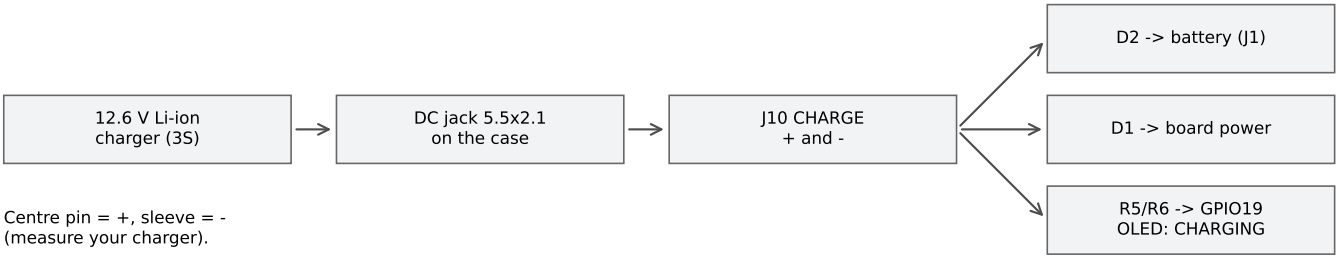
D. BENCH TEST (ThermoX_Bench.ino)

15. Plug in the ESP32 (USB end at the board edge). Upload ThermoX_Bench/ThermoX_Bench.ino. Serial Monitor 115200.
16. The OLED and the Serial Monitor show TEMP, BATT, CHARGER, B1/B2 (press each button), fan rpm, Peltier.
17. Type f: fan test. The rpm must go up at 50 % and at 100 %.
18. Connect the BTS7960 (J2, J7, J8) and the Peltier with its heat sink and fan. Type h: 10 s heat at 30 %.
    The side that should get warm gets warm. After 60 s type c: it must get cold. If it is the other way round,
    swap the two Peltier wires on M+ and M- of the BTS7960.
19. Plug the charger in: CHARGER shows IN and h / c are refused. Unplug it again.

E. NORMAL USE

20. Upload ThermoX/ThermoX.ino. For battery %: set ENABLE_BATTERY_SENSE = true and upload again.
21. Charging: SW1 to the middle, plug the 12.6 V charger into the DC jack. The OLED shows CHARGING and ThermoX
    will not run (buttons and phone too) until the charger is unplugged. Charger LED green = full.
22. Deploying all day: use USB-C (SW1 on USB, trigger board at 12 V, Peltier at 40-60 %), not the battery.
23. Never plug the USB-C charger and the DC charger in at the same time.

HOW THE CHARGER CONNECTS



Centre pin = +, sleeve = -
(measure your charger).